

Contact

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Top Skills

AI
LLM
Devin

Languages

Português (Native or Bilingual)
Alemão (Full Professional)
Inglês (Native or Bilingual)

Certifications

AWS Certified Solutions Architect – Associate
AWS Certified AI Practitioner
Microsoft Certified: Azure Fundamentals
Curso de Alemão nível B.1

Marco Menezes

Data Engineer | Data Analytics | Big Data | Azure | AWS | Databricks
Belo Horizonte, Minas Gerais, Brazil

Summary

Data Engineer with 5 years of experience in building scalable data pipelines, analytics solutions, and big data platforms. Hands-on expertise in:

- Designing and maintaining on-premises data pipelines using Cloudera (Hadoop, Hive, Spark), Linux servers (entry points, file watchers, file transfers), and BMC Control-M for orchestration.
- Re-architecting pipelines from on-premises (Cloudera / SAS) to the cloud using Azure Data Factory for ingestion and Databricks for transformation, applying the medallion architecture.
- Developing streaming pipelines to replicate transactional data from IBM DB2 to Amazon Aurora PostgreSQL using Kafka, Amazon MSK, and MSK Connect using CDC.

Experienced with CI/CD and DevOps tooling: Git, Jenkins, GitHub Actions, Jira, ServiceNow, Docker.

Strong skills with Python (10+ years), SQL (5+ years), Shell scripting (5+ years), and Scala (2+ years).

Certifications:

- AWS: Cloud Practitioner, Data Engineer Associate, and Solutions Architect Associate.
- Databricks: Apache Spark Developer and Data Engineer Associate.
- Azure: AZ-900, DP-900, SC-900, AI-900, and DP-203.
- Airflow: Airflow 2 Fundamentals.

Bachelor's degree in Control and Automation / Mechatronics Engineering obtained in August 2018. While studying complex subjects, such as algorithms, differential equations, electronics, embedded and control systems, and robotics I could learn low level stuff, and also great techniques to help me solving complex problems, very useful for data engineers.

Experience

Santander Brazil - F1rst

5 years 1 month

Senior Data Engineer

February 2024 - Present (2 years 4 months)

Belo Horizonte, Minas Gerais, Brasil

Working across Accounts & Fees (2024) and Finance (2025) business domains. Responsible for designing, developing, and maintaining data pipelines.

Key Activities:

- ✳ Migrated systems from on-premises (Cloudera and SAS/Oracle) to cloud (Azure + Databricks).
- ✳ Created and configured Azure resources including ADLS, Key Vault, Data Factory, and SPNs.
- ✳ Implemented solutions following Data Mesh (governance) and Medallion architecture guidelines.
- ✳ Migration of Accounts & Fees data pipelines to serve Open Finance requirements.
- ✳ Redesign of a streaming-only solution into real-time replication with on-demand consumption.
- ✳ Development and deployment of replication flows for core Current Account tables between IBM DB2 and Amazon Aurora:
 - [as-is]: DB2 => CDC Agent => Kafka => Spark Streaming On-Prem => [Scylla, Azure CosmosDB].
 - [to-be]: DB2 => CDC Agent => Kafka => Amazon MSK => MSK Connect => Aurora -> RESTful APIs.

Key Achievements:

- ✳ Successfully migrated batch pipelines to Azure/Databricks and streaming pipelines to AWS.
- ✳ Certified by Databricks as an Apache Spark Developer.
- ✳ Certified by AWS as Solutions Architect + Data Engineer Associate, Cloud and AI Practitioner.

Mid Level Data Engineer

July 2022 - February 2024 (1 year 8 months)

São Carlos, São Paulo, Brazil

Working in the Internal Systems business domain. Responsible for the “Operational Decommissioning Plan” and its migration to the cloud.

Activities:

- ✳ Migrated systems from on-premises (Cloudera) to Azure + Databricks.
- ✳ Created and configured ADLS, Key Vault, Data Factory, Event Hub, and SPNs.
- ✳ Used Databricks with Python, SQL, and Scala, and BMC Control-M for pipeline orchestration.
- ✳ Implemented solutions following Data Mesh (governance) and Medallion architecture guidelines.
- ✳ Extensive use of CI/CD tools including Git, Jenkins/GitHub Actions, Docker, and ServiceNow throughout the development lifecycle.

Key Achievements:

- ✳ Successfully migrated pipelines to the cloud, ensuring operational continuity, performance improvements, cost optimization, availability, and durability.
- ✳ Certified by Databricks as Data Engineer Associate.
- ✳ Certified by Microsoft as Data Engineer (DP-203).
- ✳ Certified by Microsoft as Azure+Data+AI Practitioner (AZ-900, DP-900, SC-900, and AI-900).

Data Engineer

May 2021 - July 2022 (1 year 3 months)

São Carlos, São Paulo, Brazil

Working in the Finance business domain. Junior Data Engineer with progressive responsibilities:

- ✳ Initially focused on monitoring and maintaining batch data pipelines.
- ✳ Participated in building ETL processes to generate financial and regulatory reports.
- ✳ Hands-on experience with big data tools, including Hadoop, Spark, and Hive.
- ✳ Used BMC Control-M as an orchestration tool.
- ✳ Extensive use of Python/PySpark for data processing and SQL for data validation and discovery.
- ✳ Shell scripting to create components such as File Watchers, File Transfers, and others.
- ✳ Experience with Agile methodologies (Scrum and Kanban) and CI/CD tools throughout the development lifecycle.

Key Achievements:

- ✳ Earned Advanced Data Engineer certification in Santander Brazil's Data Master program, receiving a cash award and additional benefits.

Trie Engenharia

Data Analyst Jr

November 2020 - May 2021 (7 months)

Belo Horizonte - remote, Brazil

- ✳ Brief involvement in the development and maintenance of a Python-based tool for data ingestion and processing using the Lambda architecture. The tool is aimed at serving clients in the cement industry, utilizing data generated from industrial plants and information provided by collaborators to create dashboards and reports for monitoring and Business Intelligence.

- ✳ Developed an extension for JupyterLab using Python (back-end) and React (front-end) to integrate with the OpenFOAM fluid dynamics simulation software, employed in simulating furnaces in cement industrial plants.

Colegio Técnico Inconfidentes Álvares Maciel

Adjunct Instructor

April 2015 - July 2018 (3 years 4 months)

Ouro Preto, Minas Gerais, Brazil

Instructor delivering technical courses in Industrial Mechanics, Electrotechnics, and Occupational Safety.

Key subjects taught:

- ✳ CNC Machine Programming - Theoretical foundations of computer-assisted processes and integrated manufacturing systems.
- ✳ Introduction to Hydraulics - Fundamentals and methodologies for hydraulic and pneumatic system analysis and design using FluidSIM.
- ✳ Technical Drawing II - Mechanical part design using AutoCAD.
- ✳ Introduction to Computer Science - Computer basics, operating systems, and MS Office suite (Excel, Word, PowerPoint).
- ✳ Applied Mathematics in Electrotechnics.

Universidade Federal de Ouro Preto

Laboratory Assistant

August 2016 - August 2017 (1 year 1 month)

Ouro Preto Area, Brazil

- ✱ Hardware developer (control and power electronics) and software (embedded C programming in microcontroller);

- ✱ Engaged in a project focused on the creation of a gas furnace for thermal testing, incorporating a time-based temperature control system;

Analógica Instrumentação e Controle

Summer Internship

January 2015 - February 2015 (2 months)

Belo Horizonte Area, Brazil

- ✱ Contribution to software development for data acquisition, applied to inventory management in fuel stations.

- ✱ Creation of an interface between Raspberry Pi and TLS-300 (device used for measuring parameters in fuel station tanks), utilizing the Python programming language.

Education

Federal University of Ouro Preto

Bacharelado em Engenharia, Bachelor of Engineering · (2018)